Data Science & Artificial Intelligence Training Curriculum

Program: Youth in AI × Proficio Global Impact Foundation

Target Audience: Senior High School Graduates, University Students, Young Professionals, and ToT Facilitators

Delivery Method: Virtual technical workshops, hands-on practical sessions, and group capstone projects

Phase 1:

Foundations - Data Literacy & Systems Architecture

Objective: Establish a robust baseline in computational thinking, programming, and data management for all participants, regardless of prior experience.

Module 1.1: Introduction to the AI Era

- Understanding the digital economy and the mission to democratize AI across Africa.
- Differentiating between Data Analytics, Machine Learning, and Artificial Intelligence.
- Setup of local development environments and cloud-based notebooks (e.g., Google Colab, Jupyter).

Module 1.2: Python Programming Fundamentals

- Core syntax, variables, and data structures (lists, dictionaries, tuples).
- Control flow (loops, conditionals) and algorithmic logic.
- Functions, modular programming, and debugging techniques.

Module 1.3: Database Management & SQL

- Introduction to relational database architecture.
- Writing efficient SQL queries (SELECT, JOIN, GROUP BY, subqueries).
- Data extraction and securely linking databases to local environments.

Phase 2:

Data Science - Analytics, Visualization, & Strategy

Objective: Transition from data extraction to data storytelling, equipping participants to clean, analyze, and visualize data to drive decision-making.

Module 2.1: Data Wrangling & Exploratory Data Analysis (EDA)

- Utilizing Python libraries (Pandas, NumPy) for large dataset manipulation.
- Handling missing values, outliers, and data normalization.
- Conducting EDA to identify trends and correlations.

Module 2.2: Advanced Data Visualization

- Building dynamic dashboards and reporting interfaces using Power BI.
- Principles of Premium Flat Design in data visualization (high contrast, clear data hierarchies).
- Connecting live data sources to visualization platforms.

Module 2.3: Applied Analytics (Real-World Case Studies)

- Analyzing datasets relevant to social and economic challenges in Ghana.
- Practical Application: Environmental, Social, and Governance (ESG) data analysis and foundational Carbon Accounting metrics.

Phase 3:

Machine Learning - Predictive Modeling & Cloud Computing

Objective: Move from descriptive analytics to predictive modeling, utilizing modern cloud infrastructure to train and deploy machine learning models.

Module 3.1: Machine Learning Mechanics

- Supervised vs. Unsupervised learning models.
- Feature engineering, train/test splitting, and cross-validation.
- Implementing Linear Regression, Decision Trees, and Clustering algorithms using Scikit-Learn.

Module 3.2: Cloud Infrastructure & AI Platforms

- Introduction to cloud computing architectures (AWS and Google Cloud Platform).
- Managing compute resources, cloud storage, and security protocols for AI projects.

Module 3.3: Model Deployment & MLOps

- Transitioning models from local environments to production.
- Deploying predictive models utilizing containerization and cloud services (e.g., Cloud Run, Vertex AI).
- Monitoring model performance and API integration.

Phase 4:

Advanced AI - Applied Innovation & Agentic Engineering

Objective: Empower participants to become innovators and creators by leveraging cutting-edge Large Language Models (LLMs) and advanced development methodologies.

Module 4.1: Generative AI & Large Language Models

- Understanding the architecture of LLMs (including Gemini and Vertex AI integration).
- Prompt engineering, fine-tuning, and retrieval-augmented generation (RAG).

Module 4.2: Next-Generation Development Methodologies

- Vibe Coding: Leveraging AI assistance for rapid prototyping and intuitive code generation.
- Agentic Engineering: Designing autonomous AI agents capable of multi-step problem-solving and task execution.

Module 4.3: Global AI Ethics & Standards

- Aligning local innovations with global ethical frameworks (integrating standards established by the UN ITU and Google.org).
- Addressing bias in datasets, ensuring data privacy, and building responsible AI systems.

Module 4.4: Final Capstone Project

- Collaborative group projects applying AI and data-driven approaches to solve local challenges.
- Full Software Development Life Cycle (SDLC) execution: from data collection and model training to cloud deployment and final presentation.
- Assessment & Train-the-Trainer (ToT) Integration
- To ensure the sustainability of the learning ecosystem:
- **Continuous Assessment:** Weekly practical coding challenges, SQL quizzes, and Power BI dashboard reviews.
- **ToT Milestone:** Facilitators will undergo "teach-back" sessions at the end of each phase, where they must demonstrate their technical grasp by explaining complex concepts (e.g., model deployment or SQL joins) back to the Technical Lead.
- **Project Evaluation:** Capstone projects will be graded on technical accuracy, code efficiency, user interface design, and real-world viability.